2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD) | 979-8-3315-1560-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCAD66269.2025.11241004

WROXIM: A Network-Level Simulation Platform for Wavelength-Routed Optical Networks-on-Chip

Jeng-De Chang[1], Zhidan Zheng[2], Liaoyuan Cheng[2], Liu-Xuan-Wei Zhang[1],
Tsun-Ming Tseng[2], Ing-Chao Lin[3], Ulf Schlichtmann[2]

[1]Dept. of Computer Science and Information Engineering, National Cheng Kung University, Tainan, Taiwan
[2]Dept. of Electronic Design Automation, Technical University of Munich, Munich, Germany
[3]Institute of Electronics, National Yang Ming Chiao Tung University, Hsinchu, Taiwan

p76121089@gs.ncku.edu.tw, zhidan.zheng@tum.de, liaoyuan.cheng@tum.de, p76131254@gs.ncku.edu.tw
tsun-ming.tseng@tum.de, iclin@nycu.edu.tw, ulf.schlichtmann@tum.de

***Abstract*—To meet the increasing demand for high-speed communication in many-cores systems, optical networks-on-chip (ONoCs) have gained attention for their ability to deliver low-latency and high-bandwidth data transmission. As a specific type of ONoCs, wavelength-routed ONoCs (WRONoCs) offer exclusive benefits such as collision-free and arbitration-free communication between cores. To guide the design and optimization of WRONoCs, many simulators have been developed to model WRONoC behavior at the device and circuit levels, offering detailed insights into photonic components. These tools have significantly advanced photonic design. As WRONoCs move closer to practical applications, network-level simulation becomes increasingly important for evaluating end-to-end communication behavior. Despite the need, existing simulators rarely support WRONoC modeling at the network level, leaving a critical gap in current toolchains. To address that, we propose the first network-level WRONoC simulation platform, WROXIM, adapted from an open-source NoC simulator, Noxim. It supports cycle-accurate modeling of optical communication behaviors and retains full compatibility with traditional electrical NoC simulations. To model realistic communication behaviors, WROXIM incorporates both optical and electrical components, including serializers, deserializers, waveguides, buffers, and processing elements (PEs). This modeling approach allows the simulator to capture end-to-end data movement from a PE through the optical interconnect to another PE. Given an application with multiple tasks, WROXIM supports task-driven workloads and provides key performance metrics such as latency, throughput, and energy consumption, offering a platform for design exploration and verification for WRONoCs.**

***Index Terms*—Wavelength-routed optical networks-on-chip, network-level simulation, on-chip optical interconnect, Noxim.**

I. Introduction

As the scale and complexity of data-intensive and latency-sensitive applications continue to grow, ranging from video processing to AI accelerators, the communication demand in modern many-core systems increases dramatically [1]–[3]. Conventional electrical networks-on-chip (ENoCs) exhibit limitations in keeping up with the rapid growth in on-chip communication. Optical networks-on-chip (ONoCs) have been proposed as a promising alternative for supporting high-speed communication [4]–[6]. Among all types of ONoCs, wavelength-routed optical networks-on-chip (WRONoCs) are well-known for providing arbitration-free communication [7]. In WRONoCs, a core can send data to another core at any time using a fixed and dedicated optical signal path. During the design phase, all signal paths in WRONoCs are pre-defined, which results in highly predictable and low-latency communication. Therefore, WRONoCs are considered as an appealing option for supporting on-chip communication in real-time and high-throughput computing systems.

To guide the design and optimization of WRONoCs, many simulators have been developed, covering various levels of photonic design. For example, at the device-level, tools such as Ansys Lumerical FDTD [8], COMSOL Multiphysics [9], and Synopsys RSoft Photonic Device Tools [10] provide accurate simulation of physical properties. They can model optical components such as waveguides, microring resonators (MRRs), and photodetectors, and simulate their physical behaviors and material responses. The device-level simulators can help researchers verify their design of optical components and evaluate the performance metrics without the need to manufacture the components. Besides the device-level, multiple circuit-level simulators have been developed, such as Lumerical INTERCONNECT by Ansys [11], OptSim and Photonic Design Suite by Synopsys [12], Virtuoso Photonic Extension by Cadence [13], and Tanner L-Edit Photonics by Siemens EDA [14]. Those simulators are capable of analyzing signal propagation across optical circuits and evaluating the insertion loss and other important performance factors.

Currently, the trend of WRONoC design towards is moving toward larger-scale and more complex applications with various communication scenarios. A network-level simulator enables designers to evaluate the overall communication performance of WRONoCs, including latency, throughput, scalability, and energy consumption under realistic applications. Compared to system-level simulation, network-level simula-

This work is supported in part by Bundesministeriums für Bildung und Forschung - National Science and Technology Council Taiwan (BMBF-NSTC) DE-TW-PI3D project. This work is also supported in part by the National Science and Technology Council, Taiwan under Grant NSTC 113-2221-E-006-215, NSTC 113-2923-E-006-009, NSTC 114-2923-E-006-003.

tion enables more detailed analysis of routing and flow control within the interconnnect. For example, WRONoC designs often focus on reducing transmission time and latency, but cannot observe potential queuing delays caused by buffer contention at the network-level. When multiple cores try to send data to the same destination, buffers can quickly become overwhelmed, and this effect is invisible without a network-level view. Therefore, the need to evaluate the behavior of WRONoCs at the network-level become ever-increasingly critical. However, to the best of our knowledge, the literature of network-level simulation for ONoCs, especially for WRONoCs, is markedly limited. That introduces a critical gap in current WRONoC toolchains.

To fill this gap, we present a network-level simulator for WRONoCs, **WROXIM**, adapted from an open-source NoC simulator: Noxim [15]. Our simulation platform, is built on a centralized many-core system, where a WRONoC router is placed on the center of the chip to facilitate communication among processing elements (PEs). To model realistic communication behaviors, WROXIM incorporates both optical and electrical components, including serializers, deserializers, waveguides, buffers, and PEs. To support cycle-accurate modeling of communication behaviors, we capture end-to-end data movement from a PE through the WRONoC router to another PE. Given an application with multiple tasks, WROXIM supports task-driven workloads and provides key performance metrics such as latency, throughput, and energy consumption, offering a platform for design exploration and verification for WRONoCs.

We test WROXIM for several applications and compare our simulation results to an analytical model, where the latency of modulation, transmission, and demodulation is calculated. According to the comparison results, the simulation results closely match the analytical estimates for the majority of PE-to-PE communications, which do not suffer queuing delays. However, when a PE receives data from multiple PEs at the same time, buffer contention and queuing effects lead to increased latency in WROXIM that cannot be captured by the analytical model. WROXIM can explicitly models data arrival, queuing, and supports clock domain synchronization using asynchronous FIFO buffering. This allows the simulator to preserve timing precision across heterogeneous communication paths and capture realistic latency behavior. These results highlight the importance of network-level simulation for evaluating WRONoC performance under practical conditions.

II. Preliminaries

A. WRONoCs

The growing demand for on-chip communication has increased the challenges faced by ENoCs [1]–[3], especially under aging-induced degradation [16], [17]. Recent study has shown that wavelength-routed optical networks-on-chip (WRONoCs) can support the high communication demands of modern many-core systems while offering better resilience to aging effects, thereby extending system lifetime [18].

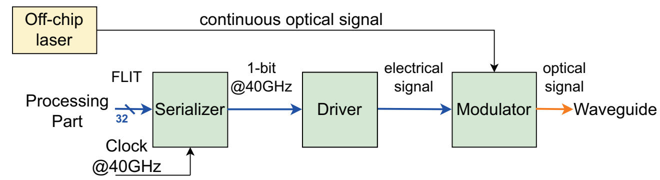


Fig. 1: The structure of a sender, which moderates signals at the frequency of $40\,\mathrm{GHz}$.

WRONoCs utilize wavelength routing to establish communication paths between PEs. Specifically, a PE, acting as a sender, sends data to another PE, acting as a receiver, using a fixed wavelength. By leveraging pre-defined wavelength channels, WRONoCs ensure predictable and high-throughput data transmission. In WROXIM, there are two main parts: the processing part and the communication part. The processing part contains processing elements and a local router. The communication part consists of three functional blocks: sending, receiving and routing. In a tile in WRONoCs, there are a PE, a local router, a sending block, and a receiving block. The routing block is shared by all tiles. Further structural and behavioral details of every part will be introduced in the following sections II-B and II-C.

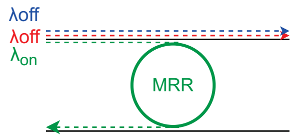
B. Communication Part

Sending block

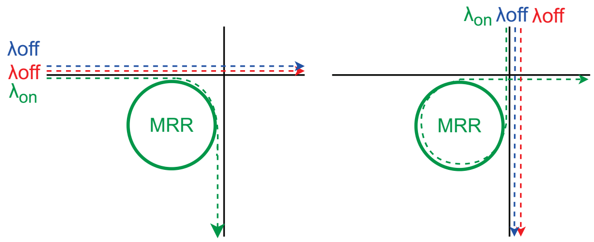
The sending part is responsible for converting electronic signals into optical signals and transmitting them to the waveguide connected to a WRONoC router. Fig. 1 shows the structure of a sender.

- Serializer: a serializer can convert an entire FLIT (flow control unit) into a single-bit stream before transmission. It sequentially outputs the data bit by bit to the driver. The serializer operates at a higher clock frequency than the system clock because the modulator can modulate the signal at a high frequency [26].
- Driver: a driver is used to amplify and condition the electrical signal to ensure proper modulation of the optical signal generated by the laser source. The driver operates with minimal delay, completing signal conversion almost instantaneously.
- Modulator: a modulator modulates an optical signal at a wavelength depending on the conditioned electrical signal from the driver. A modulator can operate at high speed, such as 10 – $40\,\mathrm{GHz}$ to support fast data modulation for WRONoC transmission [23].
- Off-chip laser: an off-chip laser can generate a continuous optical signal required for optical data transmission. Specifically, a laser supplies multiple wavelengths to support communication between a sender and a receiver. Off-chip lasers operate independently of on-chip thermal constraints, ensuring a more stable and high-power optical output [7].

Routing block



(a) Parallel switching element



(b) Crossing switching element

Fig. 2: The routing mechanism of MRRs.

The routing part ensures that optical signals are efficiently transmitted through a WRONoC router, which is formed by waveguides and microring resonators.

- Waveguide: a waveguide serves as the physical path for light propagation in optical interconnects. Multiple optical signals at various wavelengths can be transmitted in the same waveguide due to wavelength division multiplexing (WDM). The effectiveness of a waveguide depends on factors such as its length, bends, and crossing. Bends or crossings can cause signal loss and crosstalk, leading to increased energy consumption and reduced network performance.
- Microring resonator (MRR): an MRR can be used to filter optical signals based on their wavelengths. An MRR consists of a small circular waveguide and a coupling mechanism [7]. Optical signals that match the resonance wavelength of an MRR are coupled from a waveguide and redirect (drop) to another waveguide. On the other hand, signals on other wavelengths will pass through this MRR without being affected. As shown in Fig. 2, when a signal λ_{on} encounters an MRR, which is resonant to this wavelength, it will drop to the opposite/orthogonal direction (green dashed line), while the signals on λ_{off} will ignore the MRR and keep their original directions (blue and red dashed lines).
- WRONoC router: a WRONoC router is formed with waveguides and MRRs. By carefully designing the interconnect and configuring the resonance wavelengths of MRRs, a WRONoC router can establish optical paths that enable arbitration-free communication between a sender and a receiver. This routing approach ensures that optical signals are dropped at the output ports based on their assigned wavelengths, allowing concurrent transmission. WROXIM supports four state-of-the-art WRONoC routers: λ-router [19], GWOR [20], Snake [21], and Light [22].

Receiving block

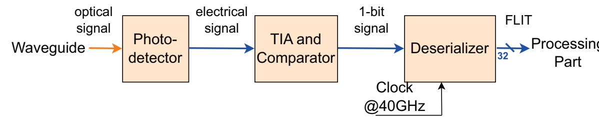


Fig. 3: The structure of a receiver, which converts optical signals back to electrical signals at the frequency of 40 GHz.

The receiving part is responsible for converting optical signals back into FLIT with parallel format.

- Photodetector: a photodetector converts incident optical signals into electrical signals through the photoelectric effect. To ensure signal integrity and minimize bit error rates, the photodetector is required to have fast temporal response and high optical sensitivity.
- Transimpedance amplifier (TIA) and Comparator: a TIA converts the weak photocurrent generated by the photodetector into a usable voltage signal. This analog voltage is then passed to a comparator, which digitizes the signal by determining whether the voltage exceeds a predefined threshold. The output of the comparator is a 1-bit digital signal. This part is designed to operate at high speed with sufficient gain to support fast and accurate signal recovery.
- Deserializer: a deserializer reconstructs the original FLIT from the serialized bit stream received from the comparator. Since WRONoCs typically transmit single bit per wavelength per clock cycle, the deserializer must reassemble the incoming bits into the parallel format to maintain proper data alignment and timing [26].

C. *Processing Part*

- Processing Element (PE): a PE is responsible for executing tasks. It generates FLITs according to the application workload and injects them into the transmission buffer for temporary storage. It also receives and processes FLITs from other PEs.
- Local router: a local router serves as the interface between the PEs and the communication part. It is responsible for handling the communication protocol and flow control. To support reliable transmission, the local router integrates a set of buffers for each sender and receiver. These buffers temporarily store FLITs and act as FIFOs to ensure synchronization across clock domains and proper coordination between transmission and reception.

D. *Communication Protocol*

In order to ensure the reliability of communication between senders and receivers, a simple and reliable communication protocol is required. In our simulator, we adopt the Alternating Bit Protocol (ABP) to handle synchronization and flow control between components. This protocol has been widely used in traditional ENoC simulators and is well-suited for modeling low-overhead point-to-point communication [24] [25].

The ABP works by assigning a 1-bit requirement (req) signal to each data unit (FLIT) and requiring an acknowledgment (ack) from the receiver before sending the next FLIT.

The control signal (req/ack) bit alternates (0↔1) between consecutive transmissions. This mechanism supports lossless and in-order delivery, while remaining simple to implement. To quantify the communication latency under ABP, we define the total communication time required to transmit a packet consisting of N FLITs between two endpoints as:

$$T_{\text{total}} = D_{\text{FLIT}} + (N-1) \times (D_{\text{FLIT}} + D_{\text{ack}}) \quad (1)$$

In Equation (1), T_{total} is the overall transmission time. D_{FLIT} is the delay for a FLIT to reach the receiver, and D_{ack} is the delay of the acknowledgment returning to the sender. This formula will be considered with the latency calculation in section IV-B.

III. Proposed Design of WROXIM

A. Development Environment

The proposed simulator, WROXIM is developed by extending Noxim [15], an open source, cycle-accurate, and network-level simulator for ENoC architectures. Noxim is implemented in SystemC, which provides both hardware, software, functionality and communication at various system levels of abstraction [26].

Noxim includes a simulation core: Noxim Runtime Engine (NRE) [15], which orchestrates the cycle-level progression of FLIT-based communication, routing logic, and buffer state updates. Building on this foundation, WROXIM extends Noxim to support wavelength-routed optical NoC architectures. While the original PE module is reused to maintain compatibility with task-graph-driven communication, and the router module is significantly redesigned to fit the design and logic of the WRONoC architectures.

B. WRONoCs Modeling

Architecture:

To support the communication requirements of WRONoCs, we modified the original mesh-based interconnection structure used in Noxim. In the base Noxim architecture, each tile is connected to its four neighboring routers through electrical links, forming a regular mesh topology. In contrast, WROXIM restructures the network into a centralized communication model, where all tiles are directly connected to a central WRONoC router. This architecture forms the foundational communication backbone of the WRONoC system. As shown in Fig. 4, the centralized WRONoC router is positioned at the center of the layout, blue waveguide paths represent the transmission paths from sender tiles to the router, while red waveguide paths represent the reception waveguides from the router to the receiver tiles.

To model WDM technology, we implement a dedicated structure in WROXIM to support multi-wavelength communication. In real WRONoC hardware, multiple data streams at different wavelengths can coexist on a shared physical waveguide without interference. However, in SystemC, a single *sc_signal* can only carry one logic value at any given time, making it unsuitable for directly modeling concurrent multi-wavelength transmission. So, we design each waveguide

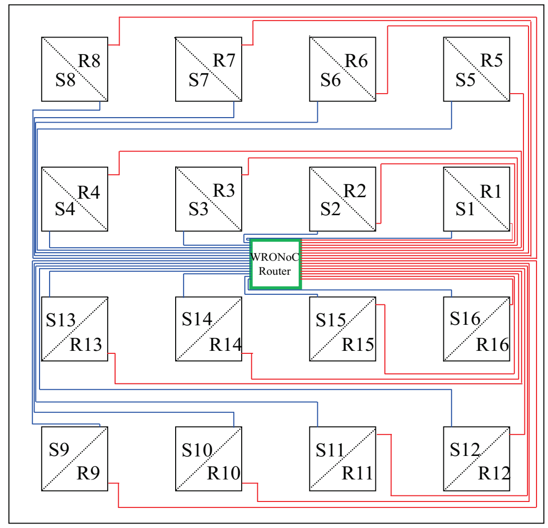


Fig. 4: The system with 16 PEs and a λ-router.

in WROXIM as a bundle of parallel signal wires, with each wire representing a dedicated wavelength channel. This design enable accurate modeling of WDM-based transmission, while maintaining accurate latency.

WRONoC router:

In the WRONoC architecture, data is transmitted along preconfigured waveguide paths through a central WRONoC router at nearly the speed of light. As described in Section II-B ***routing part***, routing in WRONoCs router is achieved through MRRs that are statically tuned to resonate with specific wavelengths. When a signal with a specific wavelength is injected, the MRR redirects it to the intended output waveguide, enabling collision-free and arbitration-free communication. To model this behavior in WROXIM, we implement the WRONoC router using a combinational logic approach. Upon injection, each serialized bit signal carries destination information, and the router immediately forwards the signal to the corresponding output tile based on its statically assigned wavelength path. This emulates the physical routing behavior in WRONoC architecture, where the signal encounters no delay within the router itself and is passively redirected by the MRRs. Since no arbitration or buffering is required within the WRONoC router, no simulation cycles are consumed during routing, preserving the cycle-accurate timing of the overall transmission.

To support various WRONoC router architectures with different wavelength assignment strategies, WROXIM adopts a configuration-based modeling framework, enabling different WRONoC architectures to be simulated by loading architecture-specific configuration files. The configuration files include the wavelength type used by each sender-receiver pair, the number of MRRs in the router, and the insertion loss of each transmisstion path. During simulation initialization, WROXIM parses these configuration files to retrieve the parameters required for power modeling. These parameters are further detailed in Section III-B ***Power modeling***. By separating architecture specification from the underlying simulation logic, this approach allows users to easily switch between WRONoC designs or experiment with custom routers without modifying internal simulation behavior.

WRONoC tile:

The WRONoC tile is the fundamental building block of the ONoC architecture, forming an n-core many-core system. As shown in Fig. 5, every tile consists of a PE for task execution, a local router that buffers data and ensures delivery reliability via the Alternating Bit Protocol, and sender/receiver modules for communication. Each sender and receiver handles a single communication direction.

Due to the higher operating frequency required by the serializer and deserializer, clock domain crossing (CDC) arises between the system clock and the optical clock domains. This is addressed by implementing asynchronous FIFO buffers between the PE and the sender/receiver modules. As shown in Fig. 5, the buffers are implemented within the local router and interface directly with the sender and receiver modules. For the sender side, the buffer is written using the same *clock_system* as the PE and read using the *clock_optical*. Conversely, on the receiver side, the buffer is written using *clock_optical* and read using *clock_system*. This clocking arrangement accurately models real-world asynchronous behavior across different clock domains.

The local router implements ABP to ensure reliable communication. To manage communication with all other tiles in the n-core NoC, the router maintains two state arrays: *current_level_tx[n]* for sending and *current_level_rx[n]* for receiving. Before transmitting a FLIT to a destination tile, the router checks whether the received ack signal from that tile has toggled (i.e., it differs from the stored value in *current_level_tx*). If the ack signal has toggled, the router proceeds to send the FLIT and updates *current_level_tx* by flipping the bit. Conversely, on the receiving side, the router monitors incoming FLITs along with their associated request bits (req). Only when the incoming request differs from *current_level_rx* and there is sufficient space in the local buffer will the FLIT be accepted, and the router responds by toggling the corresponding ack. This handshake mechanism ensures reliable and deadlock-free communication.

To manage clock domain synchronization, four separate SystemC processes are defined: *rxProcess_PE()* and *txProcess_PE()* are SC_METHODs triggered by *clock_system* for handling communication with the processing element (PE), while *rxProcess_WRONoCRouter()* and *txProcess_WRONoCRouter()* are SC_METHODs triggered by *clock_optical* for managing the interface with the WRONoC router. This separation ensures accurate timing alignment between the system and optical domains.

The sender and receiver modules are both triggered by *clock_optical*, of which only the serializer and deserializer are explicitly modeled in WROXIM, as these dominate latency in the OE/EO interface (section II-B). Other circuits such as drivers or TIA are assumed to be low-latency, non-clocked elements. While their timing behavior is not explicitly simulated, their associated power consumption is modeled and will be discussed in section III-B ***Power modeling***.

The serializer and deserializer are implemented based on the data structure proposed in [26]. Taking the serializer as an example: FLITs are stored as *sc_lv* vectors and converted into 1-bit *sc_logic* value, which are output sequentially via *bits_out* on each *clock_optical* cycle. The ABP control signals (req, ack) are appended to the end of the serialized stream to preserve accurate transmission timing. As shown in Fig. 5, the sender and receiver modules maintain flit, req and ack ports alongside *bits_out* or *bits_in*. These auxiliary ports provide necessary control and data context, allowing the WRONoC router to interpret signal destinations and simulate instant forwarding to the correct output waveguide. This implementation enables accurate simulation of high-speed photonic transmission while preserving protocol correctness, precise timing behavior, and the wavelength-based routing behavior characteristic of WRONoC architectures.

Power modeling:

On the electrical side, we adopt the original power modeling framework from Noxim for the PEs, which includes detailed estimates of dynamic and static power. For the local router, dynamic power is consumed during buffer operations such as *push* and *pop*, while static power arises from leakage and is updated at each simulation cycle. The network interface connecting the router to the PE is also modeled and contributes power only when FLITs are transferred into the PE.

$$P_{\text{laser}} = P_{\text{L}} \cdot C \cdot \lambda_{\text{act}} \tag{2}$$

$$P_{\text{Tx}} = P_{\text{drv}} \cdot \lambda_{\text{act}} + P_{\text{srl,a}} \cdot \lambda_{\text{act}} + P_{\text{srl,i}} \cdot (\lambda_{\text{tot}} - \lambda_{\text{act}}) \tag{3}$$

$$P_{\text{Rx}} = P_{\text{TIA}} \cdot \lambda_{\text{act}} + P_{\text{cmp,a}} \cdot \lambda_{\text{act}} + P_{\text{cmp,i}} \cdot (\lambda_{\text{tot}} \cdot C - \lambda_{\text{act}}) \tag{4}$$

On the optical side, WROXIM models the dynamic power for off-chip laser, senders, and receivers using Equations (2), (3), and (4) from [27]. The power parameters are based on the parameter settings proposed in [27], where active power values are set to 30.0 mW for the laser, 3.0 mW for both the serializer and the driver, 2.0mW for the TIA, 1.0 mW for the comparator, and 0.6 mW for the MRR. Additionally, idle power is considered for the serializer and comparator, with values of 1.0 mW and 0.33 mW, respectively. These components rely on precise timing control for accurate power modeling. The off-chip laser is activated when a head FLIT reaches the sender-side router and deactivated once the corresponding tail FLIT is received at the destination.

In addition to dynamic power, the system also includes static power consumption from MRRs. Multiple MRRs are present in both the WRONoC router and the sender/receiver modules. MRRs consume power continuously, regardless of transmission activity. Importantly, the number of MRRs varies depending on the router architecture, and WROXIM uses the architecture-specific configuration file (as described in section III-B **WRONoC router**) to determine the exact count of MRRs for each topology. Furthermore, insertion loss along each optical path is also considered in power estimation. The corresponding power loss is calculated using Equation (5),

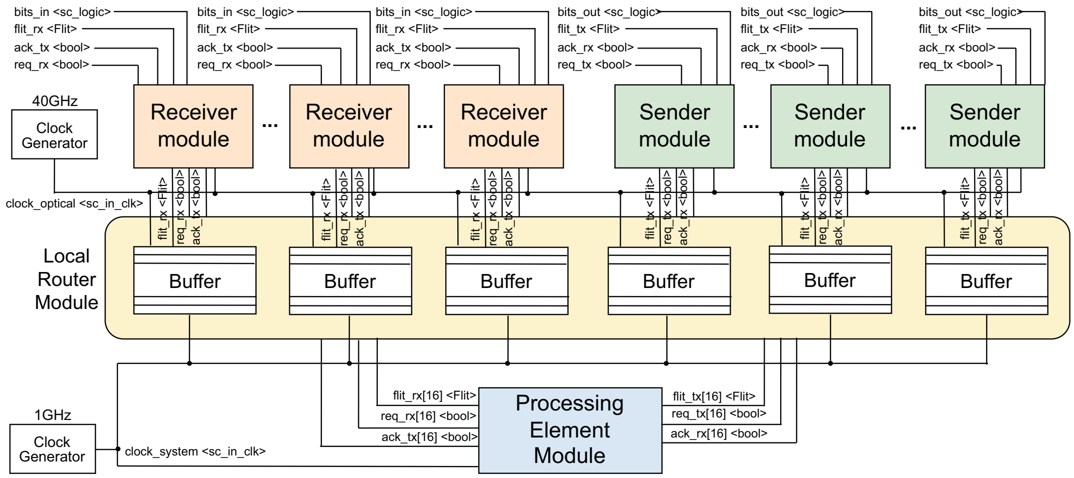


Fig. 5: A WRONoC tile module as described in SystemC.

which accounts for physical characteristics such as receiver sensitivity, coupling efficiency, and laser efficiency. In our model, the receiver sensitivity is set to –20 dBm, the coupling efficiency is 0.9, and the laser efficiency is 0.2. These values are used to compute $W_{\text{path\_loss}}$, which denotes the maximum path loss associated with a given wavelength type.

$$P_{\text{path\_loss}} = 10^{\frac{W_{\text{path\_loss}}+Er}{10}}/(Ec \times El) \tag{5}$$

IV. Case Studies

A. Experimental Setup

To validate the correctness and evaluate the performance of the WROXIM, we first applied a tool, TGFF [28] to generate four synthetic task graphs, denoted as tg0 to tg3. Each task graph models an application workload with distinct characteristics in terms of task count, execution time, and communication volume, as summarized in Table I. The tg0 is used specifically for latency correctness validation, and tg1–tg3 serve as performance evaluation workloads across different traffic intensities and task complexities. tg0 consists of a moderately sized graph (32 tasks) with medium-level communication volume (100 $\pm$ 10 FLITs), making it suitable for latency correctness validation. The remaining three task graphs, tg1, tg2, and tg3, are designed to emulate different traffic scenarios:

- tg1 models a high-parallelism, low-communication workload, characterized by a large number of tasks (73 tasks) and a small communication volume (16 $\pm$ 2 FLITs), suitable for evaluating scenarios where task-level concurrency dominates and communication overhead is minimal.
- tg2 models a medium-scale workload with moderate concurrency (39 tasks) and moderate communication volume (128 $\pm$ 16 FLITs), representing applications with balanced compute and communication characteristics.
- tg3 models a communication-intensive workload with a large data volume (1024 $\pm$ 128 FLITs) and a wide execution time range (1200-1800 cycles), suitable for stress-testing scenarios where communication cost becomes the primary performance bottleneck.

TABLE I: The information of all task graphs

Application ID	**Information**		
	Number of tasks	***Exec. time***	***Commu. volume***
tg0	32	90-110 cycles	100$\pm$10 FLITs
tg1	73	280-320 cycles	16$\pm$2 FLITs
tg2	39	600-800 cycles	128$\pm$16 FLITs
tg3	38	1200-1800 cycles	1024$\pm$128 FLITs

These three task graphs are used to evaluate the performance of different interconnection architectures, including ENoC and four WRONoCs. The evaluation focuses on three key metrics: latency, throughput, and power consumption. These benchmarks provide a diverse and representative workload set for analyzing how each architecture performs under different system conditions and traffic demands.

Next, we define the simulation environment parameters used in WROXIM. These configurations are consistently applied across all subsequent experiments to ensure fairness. The settings are summarized in Table II. The ENoC adopts a 4$\times$4 2D mesh topology, while the WRONoC employs a one of the four topologies introduced before. Both architectures are configured for a 16-core many-core system. While ENoC simulations employ the XY routing algorithm, WRONoCs rely on pre-defined wavelength-routed paths and do not require routing logic. The processing elements operate at 1 GHz, and optical sender/receiver modules run at 40 GHz. Buffer depth and FLIT size are fixed at 8 and 32 bits, respectively.

B. Latency Validation

In this part, we compare the latency with the simulation results and the analytical latency model derived by extending the transmission delay formulation presented in [29], incorporating the timing characteristics of the Alternating Bit Protocol.

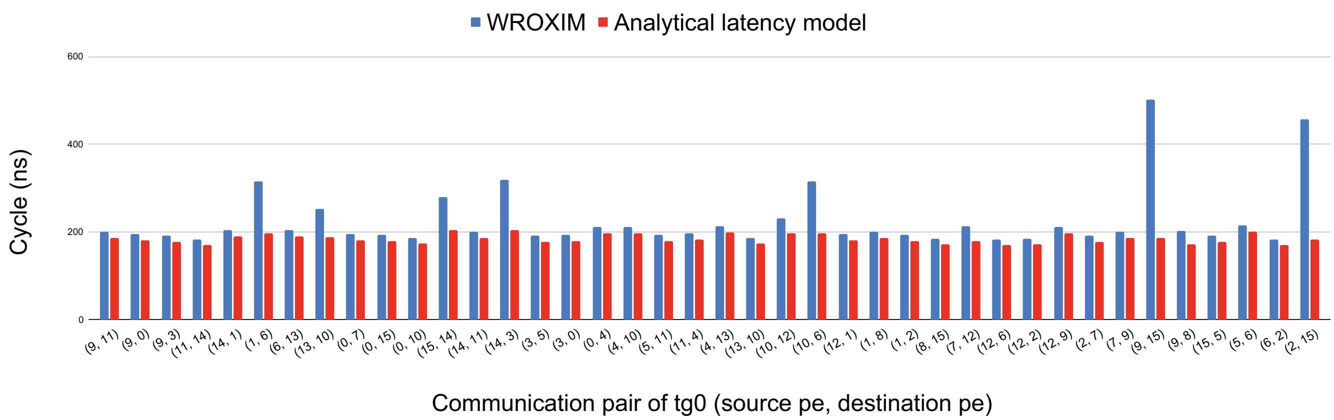


Fig. 6: Communication latency for PE-to-PE in tg0

TABLE II: Simulator Configuration Parameters

Parameter	Value
Network scale	16 cores (All architectures)
Routing algorithm	XY (ENoC only)
Buffer depth	8 FLITs (All architectures)
FLIT size	32 bits (All architectures)
System clock frequency	1 GHz (All architectures)
Sender/Receiver clock frequency	40 GHz (WRONoCs only)

To analyze the end-to-end latency of PE-to-PE communication in our WRONoC architectures, we divide the transmission process into three stages: (1) PE-to-buffer, (2) buffer-to-buffer, and (3) buffer-to-PE. Among these, both the PE-to-buffer and buffer-to-PE stages can operate concurrently with other stages during multi-FLIT transmission, effectively forming a pipeline that overlaps with the core transmission path. In contrast, the buffer-to-buffer stage is governed by the Alternating Bit Protocol, which requires each FLIT to be sent only after receiving the acknowledgment. The latency of stage (2) is calculated using Eq. 1. The total transmission latency for a packet consisting of N FLITs is estimated using the following formula:

$$T_{\text{total}} = \overbrace{D_{\text{PE-Buf}}}^{\text{stage (1)}} + \overbrace{D_{\text{Buf-Buf}} + (N-1) \times (D_{\text{Buf-Buf}} + D_{\text{ack}})}^{\text{stage (2)}} + \overbrace{D_{\text{Buf-PE}}}^{\text{stage (3)}} \tag{6}$$

In Eq. (6), $D_{\text{PE-Buf}}$ and $D_{\text{Buf-PE}}$ represent the delays between the processing element and the buffer at the sender and receiver sides, respectively. $D_{\text{Buf-Buf}}$ denotes the transmission delay between buffers. This value is derived from the optical latency formulation: $L^{\text{o}}_{\text{payload}} = W^{\text{o}}_{\text{payload}}/R_{\text{OEEO}} + dn/c$ presented in [29], which models the payload transmission delay in ONoC systems. We further extend this formulation by incorporating two additional clock cycles: one cycle for a FLIT from sender-side buffer to serializer, and another for a FLIT from deserializer to receiver-side buffer. These additions refine and complete the estimation of buffer-to-buffer communication delay in WRONoCs.

To evaluate latency accuracy, we conducted an experiment using tg0, where we extract the PE-to-PE communication latency for each task pair during their first execution, based on simulation results from WROXIM. These values are then compared against the corresponding analytical latencies calculated using the analytical model described earlier. The resulting latency differences are presented in Fig. 6, showing the latency values for each communication pair. Most cases exhibit small deviations, typically within the 7% difference. This level of discrepancy is primarily attributed to the difference in assumptions of R_{OEEO} regarding the OE/EO interface data rate used in the analytical model versus the effective transmission behavior observed in simulation. We adopted 40 Gbps for this parameter, corresponding to the serializer and deserializer operating frequency in our simulation setup. However, while the simulation clock frequency for serialization is set to 40 GHz, the actual serialization process involves an intermediate FIFO stage, where each FLIT is first buffered before being serialized. As a result, the effective data rate is slightly lower than the ideal 40 Gbps assumed in the analytical model. This mismatch between the theoretical and simulated OE/EO data rates leads to small deviations in latency estimation, which remain within an acceptable range.

However, a few communication cases show significantly larger differences. For example, task pairs (1, 6) and (10, 6) exhibit differences of approximately 120 cycles, while (9, 15) and (2, 15) show differences up to 274 and 316 cycles. These large discrepancies occur because the analytical model does not account for dynamic buffer contention, which arises when a PE simultaneously receives data from multiple sources. In such cases, buffers can become congested, causing backpressure and transmission delays that are accurately captured by the simulator, but omitted from the theoretical formula.

In addition, some task pairs such as (13, 10), (15, 14), and (14, 3) exhibit 65-116 cycles differences despite appearing conflict-free. This is due to task re-execution timing overlap, where a task's second execution begins within the same time window as other active communications targeting the same PE. As a result, previously idle paths become temporarily congested.

These discrepancies do not indicate simulation error; rather, they highlight the value of using a cycle-accurate simulator to capture real-world behaviors that are invisible to analytical models, particularly under dynamic execution and communication scenarios.

C. Performance Evaluation

For performance evaluation, we select three synthetic task graphs: tg1, tg2, and tg3, representing diverse workload scenarios in terms of parallelism and communication intensity. The evaluation focuses on three performance metrics: latency, throughput, and energy consumption for different WRONoC routers and ENoCs. We summarized the results across three key metrics:

Average latency:

As shown in Table III, WRONoCs achieve significantly lower latency compared to the ENoC across all workloads. By comparing the ENoC and WRONoCs in tg1, the average latency drops from 9.53 cycles to 2.03 cycles. In tg2 and tg3, the latency reduction is even more obvious, from 44.15 to 2.29 cycles, and from 774.35 to 2.56 cycles, respectively. This dramatic improvement demonstrates WRONoC contention-free transmission and the high-speed optical links, especially under high communication intension (as in tg3).

TABLE III: Average latency (cycle)

	ENoC	λ-router	GWOR	Snake	Light
tg1	9.53	2.03	2.03	2.03	2.03
tg2	44.15	2.29	2.29	2.29	2.29
tg3	774.35	2.56	2.56	2.56	2.56

Network throughput:

WRONoCs consistently demonstrate higher or comparable network throughput relative to the ENoC. In tg2 and tg3, which involve moderate to high communication intensity, WRONoC achieves throughput improvements from 1.03 to 1.05 FLITs/cycle and from 1.97 to 2.28 FLITs/cycle, respectively. This is mainly attributed to the dedicated and contention-free wavelength channels provided by WRONoC, enabling more efficient and parallel data transfers. In contrast, tg1 shows negligible difference in throughput across all architectures, with values remaining at approximately 0.333 FLITs/cycle. This is because tg1's overall execution time is dominated by task computation latency, rather than communication. Since network throughput is calculated as the total number of FLITs transmitted divided by the total simulation time, benchmarks where communication is not the bottleneck (such as tg1) inherently produce lower sensitivity to NoC architecture differences in this metric.

TABLE IV: Network throughput (FLITs/cycle)

	ENoC	λ-router	GWOR	Snake	Light
tg1	0.33327	0.33328	0.33328	0.33328	0.33328
tg2	1.0326	1.0511	1.0511	1.0511	1.0511
tg3	1.9658	2.28498	2.28498	2.28498	2.28498

Communication energy:

TABLE V: Total communication energy (J)

	ENoC	λ-router	GWOR	Snake	Light
tg1	2.05E-05	4.60E-04	4.90E-04	4.88E-04	4.51E-04
tg2	2.14E-05	1.10E-03	1.18E-03	1.18E-03	1.09E-03
tg3	2.27E-05	2.20E-03	2.32E-03	2.33E-03	2.19E-03

TABLE VI: Optical communication energy (J)

	λ-router	GWOR	Snake	Light
tg1	3.40E-04	3.71E-04	3.69E-04	3.32E-04
tg2	9.81E-04	1.06E-03	1.07E-03	9.74E-04
tg3	2.08E-03	2.20E-03	2.21E-03	2.07E-03

As shown in Table V, WRONoC architectures consume significantly more energy than the ENoC across all workloads. For instance, in tg1, the total communication energy increases from $2.05 \times 10^{-5}J$ to a range of $4.5 \sim 4.9 \times 10^{-4}J$ across the WRONoCs. This increase primarily results from the optical components, where power consumption is dominated by the OE/EO interfaces and laser sources. In addition, WRONoC architectures typically require more buffering resources to interface with the photonic layer, which leads to higher static power consumption from electrical buffers due to increased leakage energy. Similar trends are observed in tg2 and tg3, where total communication energy continues to rise as communication intensity increases. While buffer leakage energy itself does not scale with traffic load, higher FLIT injection rates lead to more frequent push and pop operations, increasing dynamic energy consumption within electrical buffers. To further analyze the energy consumption, we isolate the optical communication energy (Table VI), which includes contributions from laser power, sender/receiver circuit, MRRs, and insertion loss. Among the WRONoC designs, the λ-router and Light architectures exhibit slightly better energy efficiency than GWOR and Snake. This advantage arises from their symmetric wavelength design, which enables both forward and backward communication in ABP to reuse the same wavelength type. As a result, fewer laser channels are required for bidirectional transfers, significantly reducing overall laser energy consumption.

V. Conclusion and Future Work

This paper presents WROXIM, a network-level, cycle-accurate simulation platform for WRONoCs. WROXIM enables network-level performance evaluation under realistic application workloads, supporting detailed analysis of latency, throughput, and energy consumption. It supports four state-of-the-art WRONoC routers and retains the ability to simulate conventional ENoC.

Currently, WROXIM models communication as one bit per cycle per wavelength, consistent with most existing WRONoC designs. However, it has been pointed out that MRRs are capable of supporting multiple resonant wavelengths, thereby enabling bit-level parallelism to improve bandwidth. This approach may lead to increased power consumption and complexity. In future work, we plan to extend WROXIM to support such parallel transmission mechanisms and explore the trade-offs between throughput and energy efficiency in next-generation WRONoC architectures.

REFERENCES

[1] Cisco Visual Networking, "Cisco visual networking index (2017–2022)," 2019. [Online]. Available: https://www.cisco.com

[2] N. Margalit, C. Xiang, S. M. Bowers, A. Bjorlin, R. Blum, and J. E. Bowers, "Perspective on the future of silicon photonics and electronics," *Appl. Phys. Lett.*, vol. 118, no. 22, p. 220501, Jun. 2021.

[3] S. V. Winkle, A. K. Kodi, R. Bunescu, and A. Louri, "Extending the power-efficiency and performance of photonic interconnects for heterogeneous multicores with machine learning," in *Proc. IEEE Int. Symp. High Perform. Comput. Archit. (HPCA)*, Feb. 2018, pp. 480–491.

[4] S. Werner, J. Navaridas, and M. Luján, "A survey on optical network-on-chip architectures," *ACM Comput. Surv.*, vol. 50, no. 6, Dec. 2017.

[5] M. Tala, M. Castellari, M. Balboni, and D. Bertozzi, "Populating and exploring the design space of wavelength-routed optical network-on-chip topologies by leveraging the add-drop filtering primitive," in *Proc. IEEE/ACM Int. Symp. Networks-on-Chip (NoCS)*, Sep. 2016, pp. 1–8.

[6] A. Truppel, T.-M. Tseng, D. Bertozzi, J. C. Alves, and U. Schlichtmann, "PSION: Combining logical topology and physical layout optimization for wavelength-routed ONoCs," in *Proc. Int. Symp. Phys. Design (ISPD)*, Mar. 2019, pp. 49–56.

[7] T.-M. Tseng, A. Truppel, M. Li, M. Nikdast, and U. Schlichtmann, "Wavelength-routed optical NoCs: Design and EDA — State of the art and future directions," in *Proc. IEEE/ACM Int. Conf. Comput.-Aided Design (ICCAD)*, Nov. 2019, pp. 1–6.

[8] Ansys Inc., "Lumerical FDTD — Photonic simulation software," Canonsburg, PA, USA. [Online]. Available: https://www.ansys.com/products/optics/fdtd

[9] COMSOL Inc., "COMSOL Multiphysics reference manual, version 6.1," 2024. [Online]. Available: https://doc.comsol.com/6.1/doc/com.comsol.help.comsol/COMSOL_ReferenceManual.pdf

[10] Synopsys Inc., "RSoft photonic device tools," Mountain View, CA, USA. [Online]. Available: https://www.synopsys.com/optical-solutions/rsoft.html

[11] Ansys Inc., "Lumerical INTERCONNECT — Photonic integrated circuit simulation tool," Canonsburg, PA, USA. [Online]. Available: https://www.ansys.com/products/optics/interconnect

[12] Synopsys Inc., "OptSim and photonic design suite," Mountain View, CA, USA. [Online]. Available: https://www.synopsys.com/photonic-solutions/optsim.html

[13] Cadence Design Systems Inc., "Virtuoso photonic integrated circuit design platform," San Jose, CA, USA. [Online]. Available: https://www.cadence.com/zh_TW/home/tools/custom-ic-analog-rf-design/virtuoso-studio.html

[14] Siemens EDA, "Tanner L-Edit photonics — Integrated photonic IC layout editor," Wilsonville, OR, USA. [Online]. Available: https://resources.sw.siemens.com/en-US/fact-sheet-tanner-s-edit-schematic-capture/

[15] V. Catania, A. Mineo, S. Monteleone, M. Palesi, and D. Patti, "Noxim: An open, extensible and cycle-accurate network on chip simulator," in *Proc. IEEE Int. Conf. Appl.-Specific Syst., Archit. Process. (ASAP)*, Jul. 2015, pp. 162–163.

[16] C.-H. Lin, I.-C. Lin, and K.-H. Li, "TG-based technique for NBTI degradation and leakage optimization," in *Proc. IEEE/ACM Int. Symp. Low Power Electron. Design (ISLPED)*, Aug. 2011, pp. 133–138.

[17] I.-C. Lin, S.-M. Syu, and T.-Y. Ho, "NBTI tolerance and leakage reduction using gate sizing," *ACM J. Emerg. Technol. Comput. Syst.*, vol. 11, no. 1, art. 4, Sep. 2014.

[18] Z. Zheng, L. Cheng, J.-D. Chang, T.-M. Tseng, I.-C. Lin, and U. Schlichtmann, "A lifetime extension framework for communication-intensive systems based on wavelength-routed optical networks-on-chip," in *Proc. Great Lakes Symp. VLSI (GLSVLSI)*, May 2025, pp. 298–305.

[19] M. Briere *et al.*, "System level assessment of an optical NoC in an MPSoC platform," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Apr. 2007, pp. 1–6.

[20] X. Tan, M. Yang, L. Zhang, Y. Jiang, and J. Yang, "On a scalable, non-blocking optical router for photonic networks-on-chip designs," in *Proc. Symp. Photonics Optoelectron. (SOPO)*, May 2011, pp. 1–4.

[21] L. Ramini, P. Grani, S. Bartolini, and D. Bertozzi, "Contrasting wavelength-routed optical NoC topologies for power-efficient 3D-stacked multicore processors using physical-layer analysis," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2013, pp. 1589–1594.

[22] Z. Zheng, M. Li, T.-M. Tseng, and U. Schlichtmann, "Light: A scalable and efficient wavelength-routed optical networks-on-chip topology," in *Proc. Asia South Pacific Design Autom. Conf. (ASP-DAC)*, Jan. 2021, pp. 568–573.

[23] S. V. Winkle, A. K. Kodi, R. Bunescu, and A. Louri, "Extending the power-efficiency and performance of photonic interconnects for heterogeneous multicores with machine learning," in *Proc. IEEE Int. Symp. High Perform. Comput. Archit. (HPCA)*, Feb. 2018, pp. 480–491.

[24] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Pearson, 2010.

[25] N. Concer, M. Petracca, and L. P. Carloni, "Distributed flit-buffer flow control for networks-on-chip," in *Proc. Int. Conf. Hardware/Software Codesign Syst. Synth. (CODES+ISSS)*, Oct. 2008, pp. 1–6.

[26] M. Briere, E. Drouard, F. Mieyeville, D. Navarro, I. O'Connor, and F. Gaffiot, "Heterogeneous modelling of an optical network-on-chip with SystemC," in *Proc. IEEE Int. Workshop Rapid Syst. Prototyp. (RSP)*, Jun. 2005, pp. 10–16.

[27] A. Narayan, Y. Thonnart, P. Vivet, C. F. Tortolero, and A. K. Coskun, "WAVES: Wavelength selection for power-efficient 2.5D-integrated photonic NoCs," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2019, pp. 516–521.

[28] R. P. Dick, D. L. Rhodes, and W. Wolf, "TGFF: Task graphs for free," in *Proc. Int. Workshop Hardware/Software Codesign (CODES/CASHE)*, Mar. 1998, pp. 97–101.

[29] M. Li, W. Liu, L. H. K. Duong, P. Chen, L. Yang, and C. Xiao, "Contention-aware routing for thermal-reliable optical networks-on-chip," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 40, no. 2, pp. 260–273, Feb. 2021.